

**AIMS**African Institute for
Mathematical Sciences
RWANDA

Lecture 3: Probability III

Introduction to Probability and Statistics

evans.gouno@univ-ubs.fr

- 1 Continuous r.v.
- 2 Distribution Functions
- 3 Moments

Continuous random variables

- X : Lap time for a given operation
- $\Omega = \mathbb{R}^+$.
- A possible outcome is: 1h, 27 min., 30 sec., 5/10 msec., ...
- In this case, it is not possible to associate an **exact** value to the outcome.
- Event: $X^{-1}(]a, b[) = \{\omega \in \Omega \mid a < X(\omega) < b\}$.
- The distribution is characterized by:

$$P\{X^{-1}(]a, b[)\} = P\{a < X(\omega) < b\} = P\{a < X < b\}.$$

Distribution function

- When a random variable is continuous, we consider:

$$P(x < X \leq x + h) = P(X \leq x + h) - P(X \leq x).$$

We define the **cumulative distribution function** (c.d.f.).

Definition

The **cumulative distribution function** is the function $F_X : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F_X(x) = P(X \leq x).$$

Cumulative distribution functions

Theorem

A function F mapping the real line to $[0, 1]$ is a **cumulative distribution function (c.d.f.)** for some probability P if and only if F satisfies the following three conditions:

- ① F is non-decreasing: $x_1 < x_2$ implies that $F(x_1) \leq F(x_2)$.
- ② F is normalized:

$$\lim_{x \rightarrow -\infty} F(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow +\infty} F(x) = 1.$$

- ③ F is right-continuous:

$$F(x) = \lim_{\substack{y \rightarrow x \\ y > x}} F(y).$$

Probability density function

- We define $f_X(x)$, the **probability density function (p.d.f.)** of X as:

$$f_X(x) = \lim_{h \rightarrow 0} \frac{1}{h} (F(x+h) - F(x)).$$

- The p.d.f. is the derivative of F_X , the c.d.f. :

$$f_X(x) = F'(x).$$

- We have:

$$F(x) = P(X \leq x) = \int_{\{y \leq x\}} f_X(y) dy \text{ and } \int_V f_X(y) dy = 1.$$

Remark: Discrete random variable

$$F(x) = P(X \leq x) = \sum_{\{y \in V | y \leq x\}} P(X = y).$$

C.d.f of a discrete r.v.

Exercise

Represente the distribution and the c.d.f of X , sum of the dots obtained rolling two dice.

x	2	3	4	5	6
$P(X = x)$	1/36	1/18	1/12	1/9	5/36

x	7	8	9	10	11	12
$P(X = x)$	1/6	5/36	1/9	1/12	1/18	1/36

Download R: <https://cran.r-project.org/bin/windows/base/> or use Python.

C.d.f of a discrete r.v.

Code R

```

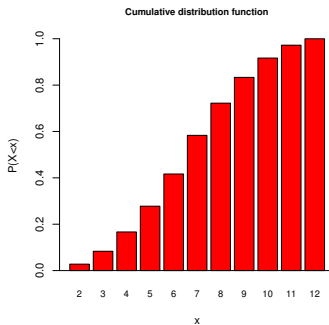
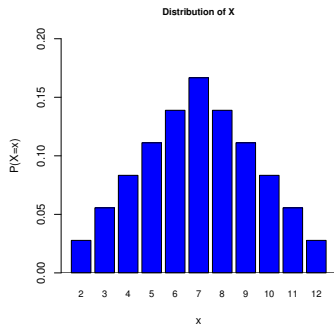
x<-seq(2,12,1)
Px<-c(1/36, 1/18, 1/12, 1/9, 5/36, 1/6, 5/36, 1/9, 1/12 ,1/18 ,1/36)

# Distribution of X
barplot(Px, col="blue", ylim=c(0,0.2), names.arg=c(as.character(x)),
cex.names=0.8,
xlab="x", ylab="P(X=x)", main="Distribution of X", cex.main=0.8)
abline(h=0)

# Cumulative distribution function
CdfX<-cumsum(Px)
barplot(CdfX, col="red", names.arg=c(as.character(x)), cex.names=0.8,
xlab="x", ylab="P(X<=x)", main="Cumulative distribution function",
cex.main=0.8)
abline(h=0)

```

C.d.f of a discrete r.v.



Examples of discrete r.v.'s distribution

Definition

A r.v. X is said to be a **Bernoulli's** r.v. with parameter p iff

$$P(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}, \quad 0 < p < 1.$$

Notation: $X \sim \mathcal{B}(1, p)$.

Definition

A r.v. X follows a **binomiale** distribution with parameters (n, p) iff

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x \in \{0, \dots, n\}, \quad 0 < p < 1.$$

Notation: $X \sim \mathcal{B}(n, p)$.

Example of continuous r.v. distribution

Definition

A r.v. X has an **exponential** distribution with parameter $\theta > 0$ iff its probability density function (p.d.f) has the form:

$$f(x | \theta) = \frac{1}{\theta} \exp \left\{ -\frac{x}{\theta} \right\}, \quad x \in \mathbb{R}^+.$$

Notation: $X \sim \mathcal{E}(\theta)$

Exercise :

- 1 Check that we have a density.
- 2 Compute the cumulative distribution function.

Example of continuous r.v. distribution

Definition

A r.v. X has an **exponential** distribution with parameter $\theta > 0$ iff its probability density function (p.d.f) has the form:

$$f(x | \theta) = \frac{1}{\theta} \exp \left\{ -\frac{x}{\theta} \right\}, \quad x \in \mathbb{R}^+.$$

Notation: $X \sim \mathcal{E}(\theta)$

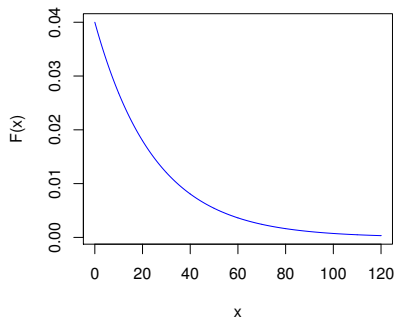
Exercise :

- 1 Check that we have a density.
- 2 Compute the cumulative distribution function.

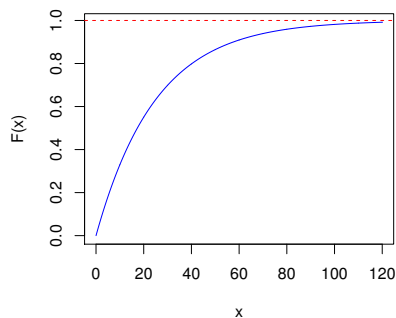
Solution: $F(x) = 1 - e^{-x/\theta}$.

Distribution of continuous r.v.'s

P.d.f of an exponential r.v



C.d.f of an exponential r.v



Expectation

Definition

The **expected value** or **mean** or **first moment** of a random variable X is defined as:

$$E(X) = \begin{cases} \sum_{x \in V} x P(X = x) & \text{if } X \text{ is discrete,} \\ \int_V x f_X(x) dx & \text{if } X \text{ is continuous.} \end{cases}$$

Example:

- $X \sim \mathcal{B}(1, p)$.
- $X \sim \mathcal{E}(\theta)$.

Moments

- Let g be a regular function. Then:

$$E[g(X)] = \begin{cases} \sum_{x \in V} g(x) P(X = x) & \text{if } X \text{ is discrete} \\ \int_V g(x) f_X(x) dx & \text{if } X \text{ is continuous.} \end{cases}$$

- q^{th} -moments

$$E(X^q) = \begin{cases} \sum_{x \in V} x^q P(X = x) & \text{if } X \text{ is discrete,} \\ \int_V x^q f_X(x) dx & \text{if } X \text{ is continuous.} \end{cases}$$

m.g.f.

Moment generating function (m.g.f)

$$M_X(t) = E[e^{tX}].$$

Lemma

$$E(X^q) = M_X^{(q)}(0).$$

➡ The m.g.f identifies the distribution.

Expectation

Properties

- $\forall b \in \mathbb{R}, E(X + b) = E(X) + b.$
- $\forall a \in \mathbb{R}, E(aX) = aEX.$

Exercise

A fair coin is tossed n times. Let X be the number of TAIL.

X is a Binomiale distribution with parameters (n, p) with $p = 1/2$

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, x = 0, 1, \dots, n.$$

Let a be a real number. Consider the random variable $Y = \frac{a^X}{2^n}$.
Compute $E(Y)$.

Expectation

Solution:

By definition, $E(a^X/2^n) = \sum_{x=0}^n (a^x/2^n)P(X=x)$.

Hence

$$\begin{aligned} E(Y) &= \frac{1}{2^n} \sum_{x=0}^n a^x \binom{n}{x} \left(\frac{1}{2}\right)^x \left(1 - \frac{1}{2}\right)^{n-x} \\ &= \frac{1}{2^n} \sum_{x=0}^n a^x \binom{n}{x} \left(\frac{1}{2}\right)^n = \frac{1}{2^{2n}} \sum_{x=0}^n \binom{n}{x} a^x = \left(\frac{a+1}{2}\right)^n \end{aligned}$$

Reminder:

Newton's binomial formula : $(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$.

Variance of a random variable

Definition

The **variance** of a random variable is defined as:

$$\text{Var}(X) = E[(X - E(X))^2].$$

$$\text{Var}(X) = \begin{cases} \sum_{x \in V} (x - E(X))^2 P(X = x) & \text{if } X \text{ is discrete} \\ \int_V (x - E(X))^2 f_X(x) dx & \text{if } X \text{ is continuous.} \end{cases}$$

Variance of a random variable

Exercise: Compute the variance for:

- $X \sim \mathcal{B}(1, p)$.
- $X \sim \mathcal{E}(\theta)$.

Variance of a random variable

Properties

- $\text{Var}(X) = E(X^2) - (EX)^2$.
- $\forall b \in \mathbb{R}, \text{Var}(X + b) = \text{Var}(X)$.
- The variance is **quadratic** : $\forall a, \text{Var}(aX) = a^2 \text{Var}(X)$.